

***COURSE: CHE 312: Transport Phenomena II***

**Spring Semester 2020**

**Department of Chemical Engineering**

**University of Illinois at Chicago**

**Chicago, Illinois 60607-7052**

**Quiz #2**

**February 12, 2020**

***Closed Notes and Book***

**Duration: 25 minutes**

A nuclear reactor fuel element consists of a solid cylindrical pin of radius  $r_f$  and thermal conductivity  $k_f$ . The fuel pin is in good contact with a cladding material of outer radius  $r_c$  and thermal conductivity  $k_c$ . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate  $\dot{q}$  and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature  $T_\infty$  and a convection coefficient  $h$ .

- a) Obtain equations for the temperature distributions  $T_f(r)$  and  $T_c(r)$  in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- b) Consider a uranium oxide fuel pin for which  $k_f = 3 \text{ W/m}\cdot\text{K}$  and  $r_f = 6 \text{ mm}$  and cladding for which  $k_c = 25 \text{ W/m}\cdot\text{K}$  and  $r_c = 9 \text{ mm}$ . If  $\dot{q} = 2 \times 10^8 \text{ W/m}^3$ ,  $h = 2000 \text{ W/m}^2\cdot\text{K}$ , and  $T_\infty = 300 \text{ K}$ , what is the maximum temperature in the fuel element?

